

Day 19: DTP / VTP

DTP (Dynamic Trunking Protocol)

- Protocol that allows SWITCHES to negotiate the status of their SWITCHPORTS, without manual configuration, to be:
 - ACCESS PORTS
 - TRUNK PORTS
- DTP is ENABLED by default on all Cisco SWITCH interfaces

We've been manually configuring SWITCHPORTS using :

- "switchport mode access"
- "switchport mode trunk"

 'show interfaces <interface-id> switchport' will show you a switchport's settings.

For security purposes, **manual configuration** is recommended. DTP should be disabled on ALL SWITCHPORTS



DYNAMIC DESIRABLE:

- This MODE will actively try to form a TRUNK with other Cisco SWITCHES.
- Will form a TRUNK if connected to another SWITCHPORT in the following modes:
 - "switchport mode trunk"
 - "switchport mode dynamic desirable"
 - "switchport mode dynamic auto"

HOWEVER ... if the other interface is set to "static access" (ACCESS mode), it will NOT form a TRUNK, it will be an ACCESS PORT

DYNAMIC AUTO:

- This MODE will NOT actively try to form a TRUNK with other Cisco SWITCHES
- Will form a TRUNK if connected SWITCH is actively trying to form a TRUNK.
- It will form a TRUNK with a SWITCHPORT in the following modes:
 - "switchport mode trunk"
 - "switchport mode dynamic desirable"

TRUNK to ACCESS connection will operate in a **Mismatched Mode**.

This configuration does NOT work and SHOULD result in an error. Traffic will NOT work.

TABLE SHOWING THE DIFFERENT MODES AND COMPATIBILITY IN FORMING A TRUNK



DTP will NOT form a TRUNK with:

a ROUTER

a PC

etcetera ...

The SWITCHPORT will be in ACCESS Mode only!

OLD SWITCHES:

- "switchport mode dynamic desirable" = Default administrative mode.

NEWER SWITCHES:

- "switchport mode dynamic auto" = Default administrative mode.

HOW TO DISABLE DTP NEGOTIATION ON AN INTERFACE:

- "switchport nonegotiate"
- "switchport mode access"

It is a security recommendation to disable DTP on all SWITCHPORTS and manually configure them as ACCESS or TRUNK ports.

ENCAPSULATION:

SWITCHES that support both:

- 802.1Q
- ISL

TRUNK encapsulation can use DTP to negotiate the encapsulation they will use.

- Negotiation is Enabled by default

```
💡 'switchport trunk encapsulation negotiate'
```

- ISL is favored over 802.1Q
 - If BOTH SWITCHES support ISL, ISL will be selected.
- DTP frames are sent in:
 - VLAN1 when using ISL
 - Native VLAN when using 802.1Q (the default native VLAN is VLAN1, however)

VTP (VLAN Trunking Protocol)

In Privileged EXEC mode:

```
💡 #show vtp status
```

- Protocol for configuring VLANs on a Central SWITCH
 - A SERVER that other SWITCHES synch. to (auto configuring by connection)
- Other switches (VTP CLIENTS) will synchronize their VLAN database to the SERVER
- Designed for large networks with many VLANs (reduces manual configuration)
- RARELY used. Recommended you DO NOT USE it
- There are THREE VTP Versions :

- v1
 - Does NOT supports Extended VLAN Range 1006-4094
 - v2
 - Does NOT supports Extended VLAN Range 1006-4094
 - Supports Token Ring VLANs ; otherwise similar to V1
 - v3
 - Supports Extended VLAN Range 1006-4094
 - CLIENTS store VLAN dBase in NVRAM
 - There are **THREE VTP modes**:
 - SERVER
 - CLIENT
 - TRANSPARENT
 - Cisco SWITCHES operate in VTP SERVER mode, by default
-



VTP SERVERS:

- Can ADD / MODIFY / DELETE VLANs
- Store the VLAN dBase in NVRAM
- Increase Revision Number every time VLAN is Added / Modified / Deleted
- Advertises **Latest Version** of VLAN dBase on TRUNK interfaces.
- VTP CLIENTS synchronize their VLAN dBase to it
- **VTP SERVERS also function as VTP CLIENTS**
 - **THEREFORE, a VTP SERVER will synchronize to another VTP SERVER with a higher Revision Number**



One danger of VTP: Connecting an old SWITCH with higher Revision Number to network (and if the VTP Domain Name matches), all SWITCHES in Domain will

synchronize their VLAN dBase to SWITCH

VTP CLIENTS:

```
💡 (config)# vtp mode client
```

- Cannot Add / Modify / Delete VLANs
- Does NOT store the VLAN database in NVRAM
 - **VTP v3 CLIENTS DO**
- Will synchronize their VLAN dBase to the SERVER with the highest version number in their VTP Domain
- Advertise their VLAN dBase and forward VTP Advertisements to other CLIENTS over TRUNK Ports

VTP TRANSPARENT MODE:

```
💡 (config)# vtp mode transparent
```

- Does NOT participate in VTP Domain (does NOT sync VLAN database)
- Maintains own VLAN dBase in NVRAM.
- Can Add / Modify / Delete VLANs
- Won't Advertise to other SWITCHES
- Will forward VTP advertisements to SWITCHES in the same Domain as it.

VTP DOMAINS

If a SWITCH with no VTP Domain (Domain NULL) receives a VTP advertisement with a VTP Domain name, it will automatically join that VTP Domain

If a SWITCH receives a VTP advertisement in the same VTP domain with a higher revision number, it will update it's VLAN database to match

REVISION NUMBERS:

There are TWO ways to RESET a REVISION NUMBER to 0:

- Change VTP Domain to an unused Domain
 - Change VTP mode to TRANSPARENT
-

VTP VERSION NUMBER

```
💡 (config)#vtp version <version number>
```

Changing the Version # will force sync/update all connected SWITCHES to the latest Version #